

Announcement

Homework #2 was posted online.

Homework #2 is due 1:15pm, Tuesday, Feb. 10.

Announcement

Quiz #1

Time and Date: 2:15pm – 2:30pm, Tuesday, Feb. 3

Topic: Histogram processing

Open book and open notes and you can use a calculator

Today's Agenda

- **Histogram processing**
 - Histogram equalization
 - Histogram matching
- **Spatial filtering**
 - Linear filters
 - Image Smoothing

Local Histogram Processing

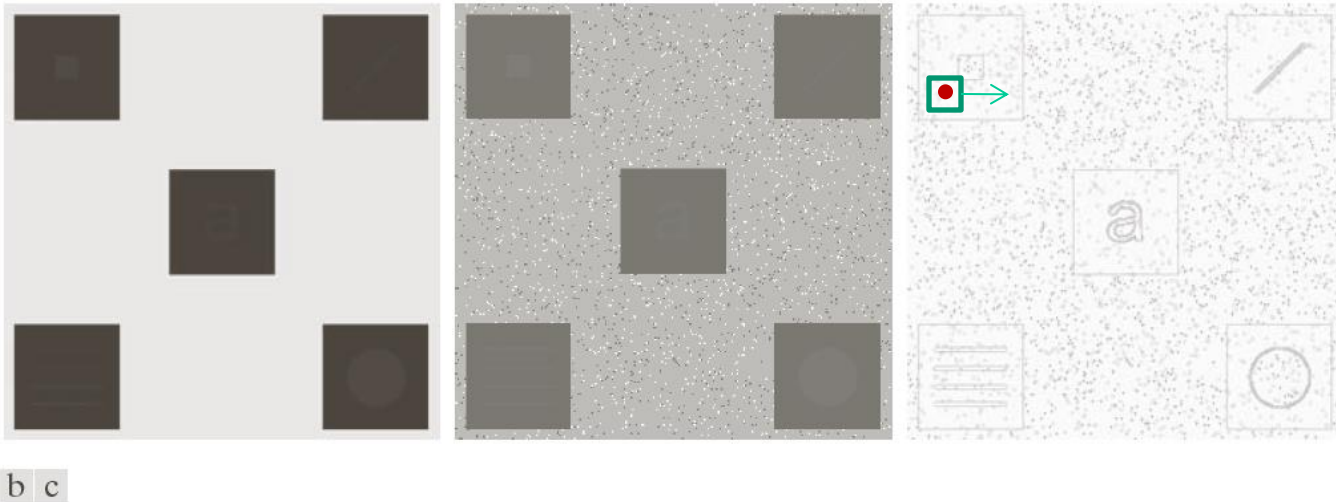
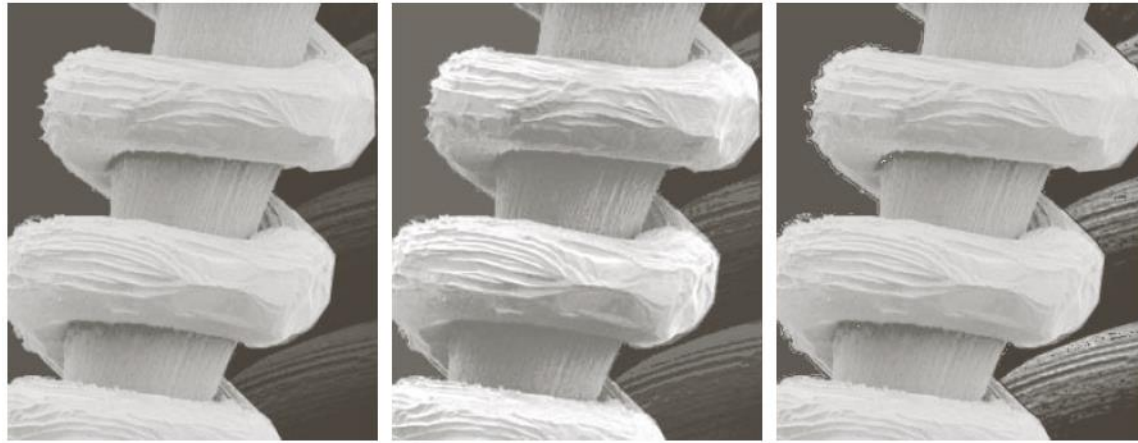


FIGURE 3.26 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization applied to (a), using a neighborhood of size 3×3 .

Using Histogram Statistics for Image Enhancement



a b c

FIGURE 3.27 (a) SEM image of a tungsten filament magnified approximately 130×. (b) Result of global histogram equalization. (c) Image enhanced using local histogram statistics. (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)

$$m_G = \sum_{i=0}^{L-1} r_i p(r_i),$$

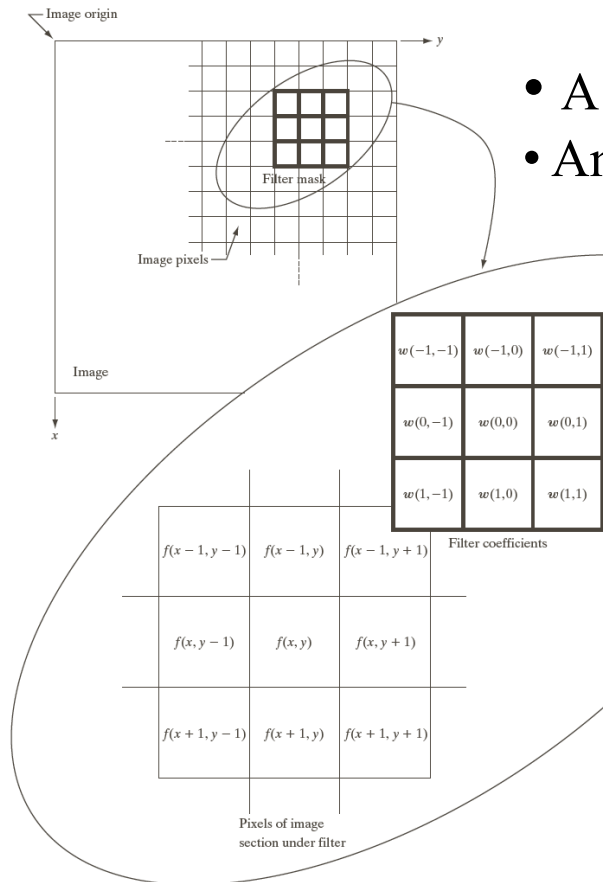
$$\sigma_G^2 = \sum_{i=0}^{L-1} (r_i - m)^2 p(r_i)$$

$$m_{S_{xy}} = \sum_{i=0}^{L-1} r_i p_{S_{xy}}(r_i),$$

$$\sigma_{S_{xy}}^2 = \sum_{i=0}^{L-1} (r_i - m_{S_{xy}})^2 p_{S_{xy}}(r_i)$$

$$g(x, y) = \begin{cases} 4f(x, y) & \text{if } m_{S_{xy}} \leq 0.4m_G \text{ AND } 0.02\sigma_G \leq \sigma_{S_{xy}} \leq 0.4\sigma_G \\ f(x, y) & \text{otherwise} \end{cases}$$

Fundamentals of Spatial Filtering



- A neighborhood
- An operator with the same size: linear/nonlinear

Note: Each element in w will visit every pixel in the image just once.

Linear spatial filtering:

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

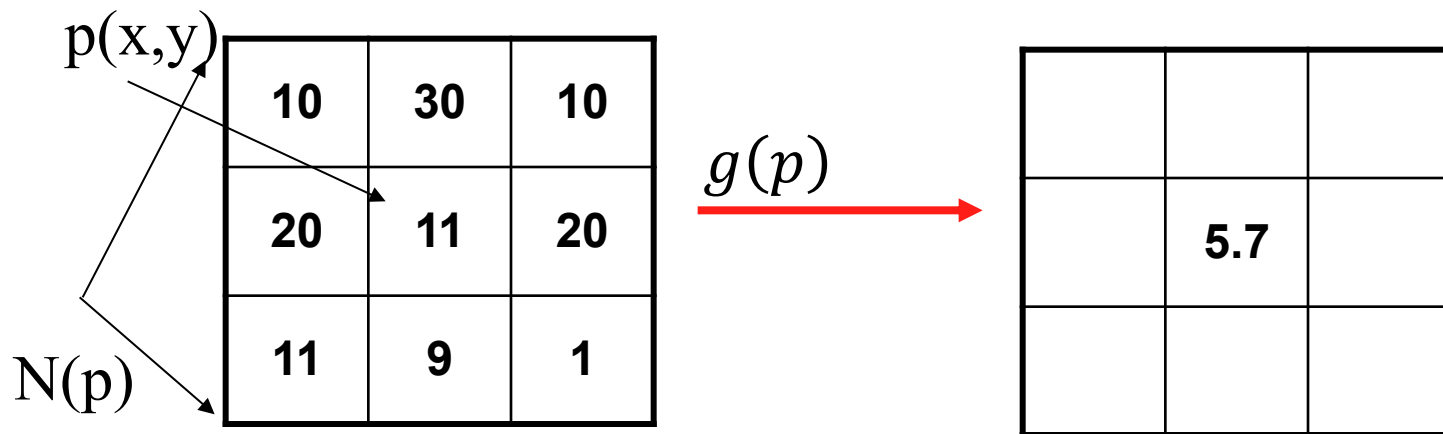
↓

$$\text{Inner product } g(x, y) = \mathbf{w} \bullet \mathbf{f} = \mathbf{w}^T \mathbf{f}$$

FIGURE 3.28 The mechanics of linear spatial filtering using a 3×3 filter mask. The form chosen to denote the coordinates of the filter mask coefficients simplifies writing expressions for linear filtering.

Fundamentals of Spatial Filtering

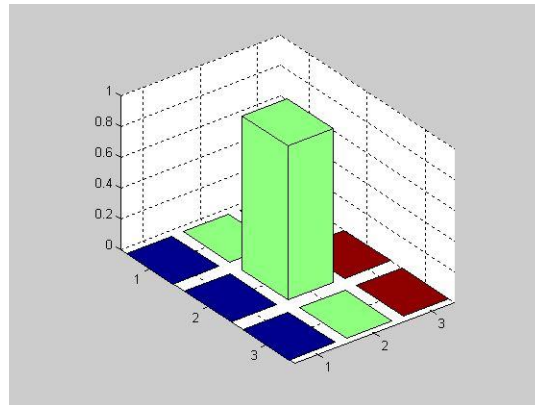
Modifying the pixels in an image based on some function of a local neighborhood of the pixels



$g(p)$:

- Linear function
 - Correlation
 - Convolution
- Nonlinear function
 - Order statistic (median)

Linear Filtering



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0	0	0
0	1	0
0	0	0

=



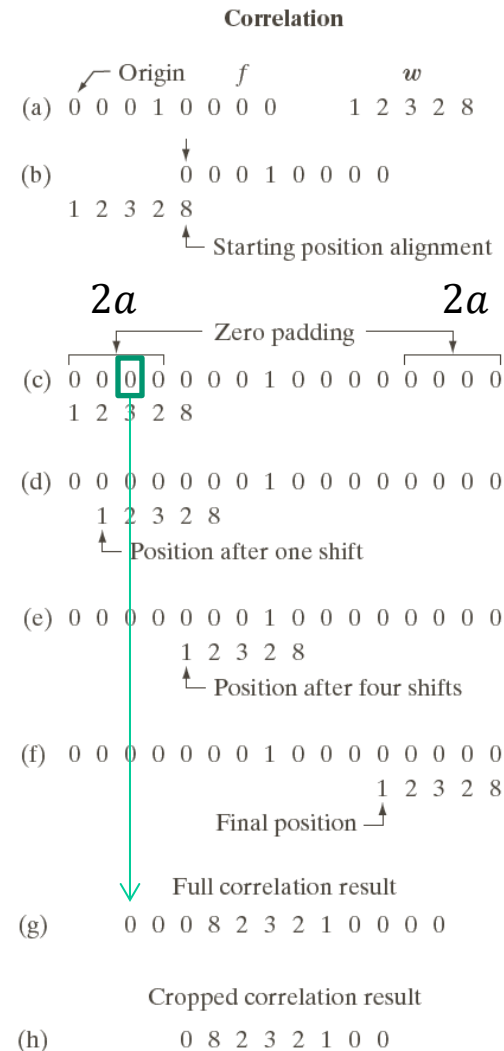
Spatial Correlation: 1D Signal

$$\text{1D correlation} \sum_{s=-a}^a w(s) f(x+s)$$

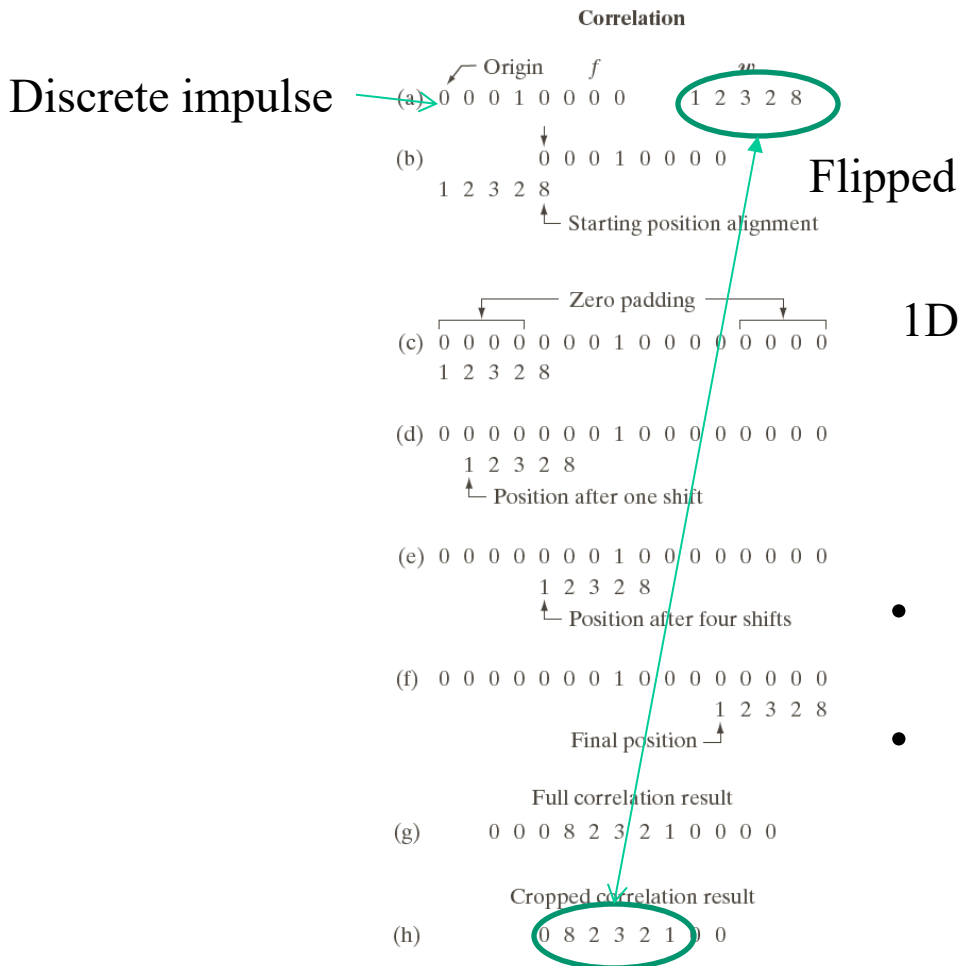
Zero-padding: add zeros on the left and right margin, respectively



- **Full correlation result** has the size of $M + 2a$
- **Cropped result** has the size of M – the size of the original signal



Spatial Correlation: 1D Signal

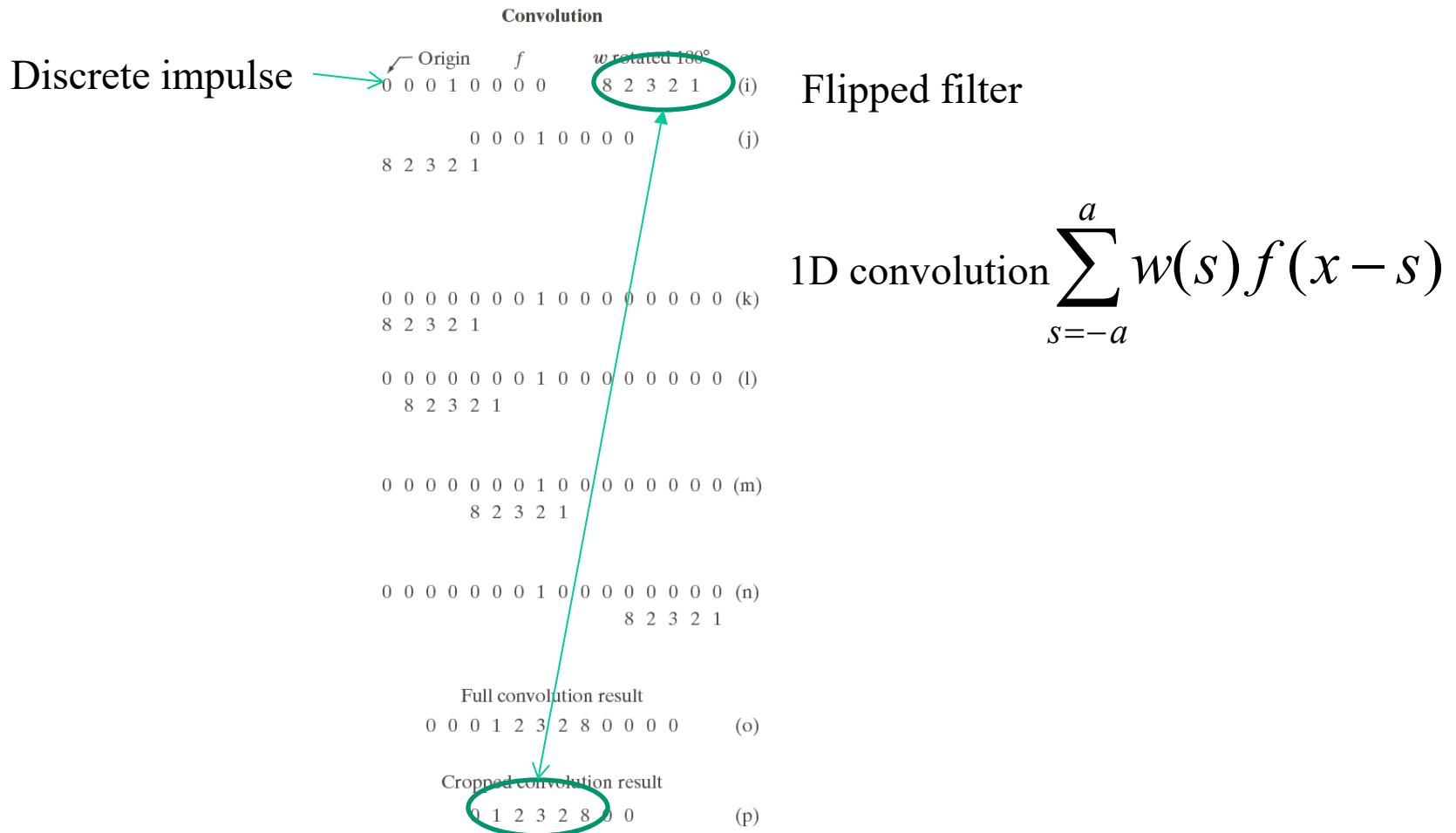


$$\text{1D correlation} \sum_{s=-a}^a w(s) f(x+s)$$

- Full correlation result has the size of $M + 2a$
- Cropped result has the size of M – the size of the original signal

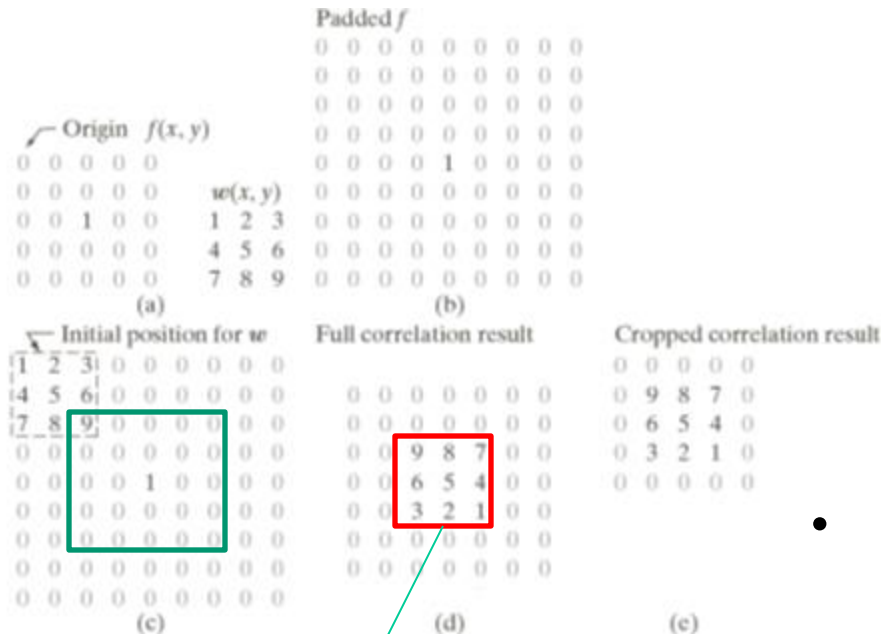
The impulse response is a rotation of the filter by 180 degree

Spatial Convolution: 1D Signal



The impulse response is the same as the filter

Extend to 2D Image: 2D Image Correlation

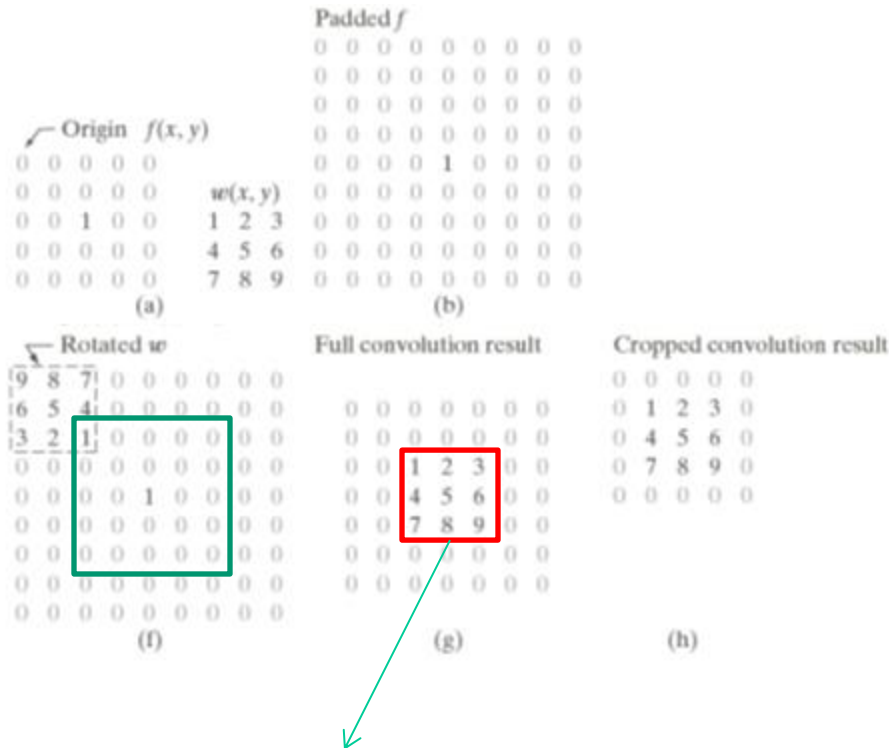


$$\sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

- Full correlation result has the size of $(M + 2a, N + 2b)$
- Cropped result has the size of (M, N) – the size of the original image

The 2D impulse response of image correlation is a rotation of the filter by 180 degree

Extend to 2D Image: 2D Image Convolution



$$\sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x-s, y-t)$$

- Flip in both horizontal and vertical directions (rotate 180 degree) -> same if the filter is symmetric
- Convolution filter/mask/kernel
- Full convolution result has the size of $(M + 2a, N + 2b)$
- Cropped result has the size of (M, N) – the size of the original image

The 2D impulse response of image convolution is the same as the filter

Linear Filters

General process:

- Form new image whose pixels are a weighted sum of original pixel values, using the same set of weights at each point.

Properties

- Output is a linear function of the input
- Output is a shift-invariant function of the input (i.e. shift the input image two pixels to the left, the output is shifted two pixels to the left)

Example: smoothing by averaging

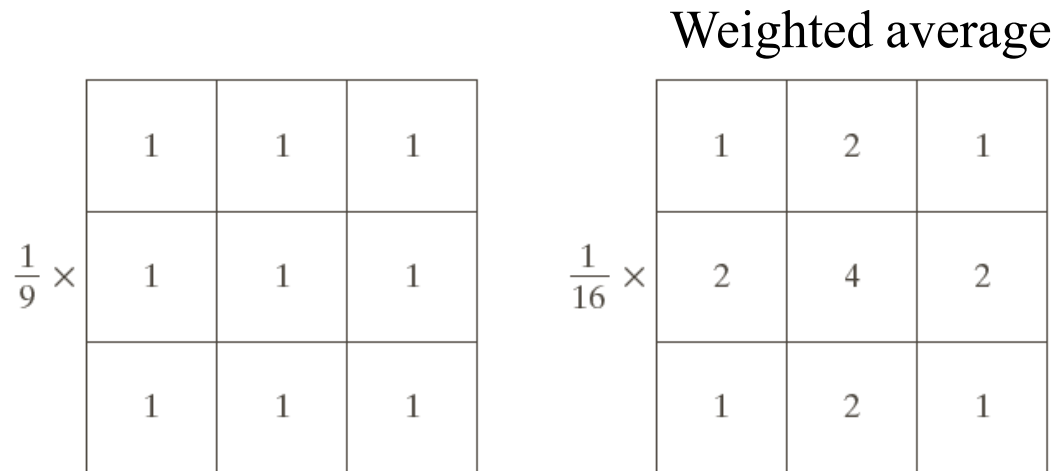
- form the average of pixels in a neighborhood

Example: smoothing with a Gaussian

- form a weighted average of pixels in a neighborhood

Example: finding an edge

Smoothing Spatial Filter – Low Pass Filters



a b

FIGURE 3.32 Two 3×3 smoothing (averaging) filter masks. The constant multiplier in front of each mask is equal to 1 divided by the sum of the values of its coefficients, as is required to compute an average.

$$g(x, y) = \frac{\sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)}{\sum_{s=-a}^a \sum_{t=-b}^b w(s, t)}$$

- Noise deduction
- reduction of “irrelevant details”
- edge blurred

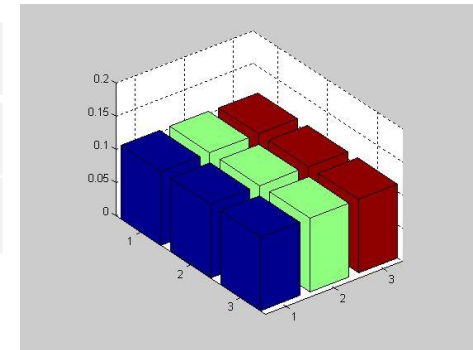
Normalization factor

Smoothing Spatial Filter

Image averaging

$$R = \frac{1}{9} \sum_{i=1}^9 z_i$$

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9



$\ast \frac{1}{9}$

1	1	1
1	1	1
1	1	1

=



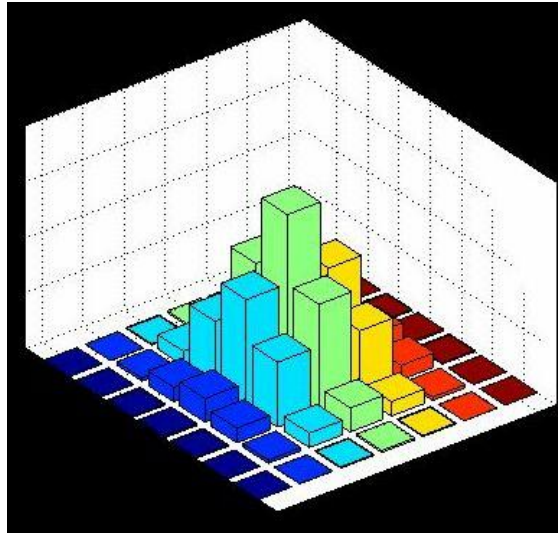
Smoothing Spatial Filter

2D Gaussian filter

$$h(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$



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Comparison using Different Smoothing Filters – Different Kernels



Average



Gaussian

Comparison using Different Smoothing Filters: Different Size

Filter size: 3, 5, 9, 15, 35

Square size: 3, 5, 9, 15, 25, 35, 45, 55
with a spacing of 25

Bar: 5x100 with a spacing of 20

Letter size: 10, 12, 14, 16, 18, 20, 24

FIGURE 3.33 (a) Original image, of size 500×500 pixels. (b)–(f) Results of smoothing with square averaging filter masks of sizes $m = 3, 5, 9, 15$, and 35 , respectively. The black squares at the top are of sizes 3, 5, 9, 15, 25, 35, 45, and 55 pixels, respectively; their borders are 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in increments of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels wide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 pixels, and their borders are 15 pixels apart; their intensity levels range from 0% to 100% black in increments of 20%. The background of the image is 10% black. The noisy rectangles are of size 50×120 pixels.

a b
c d
e f

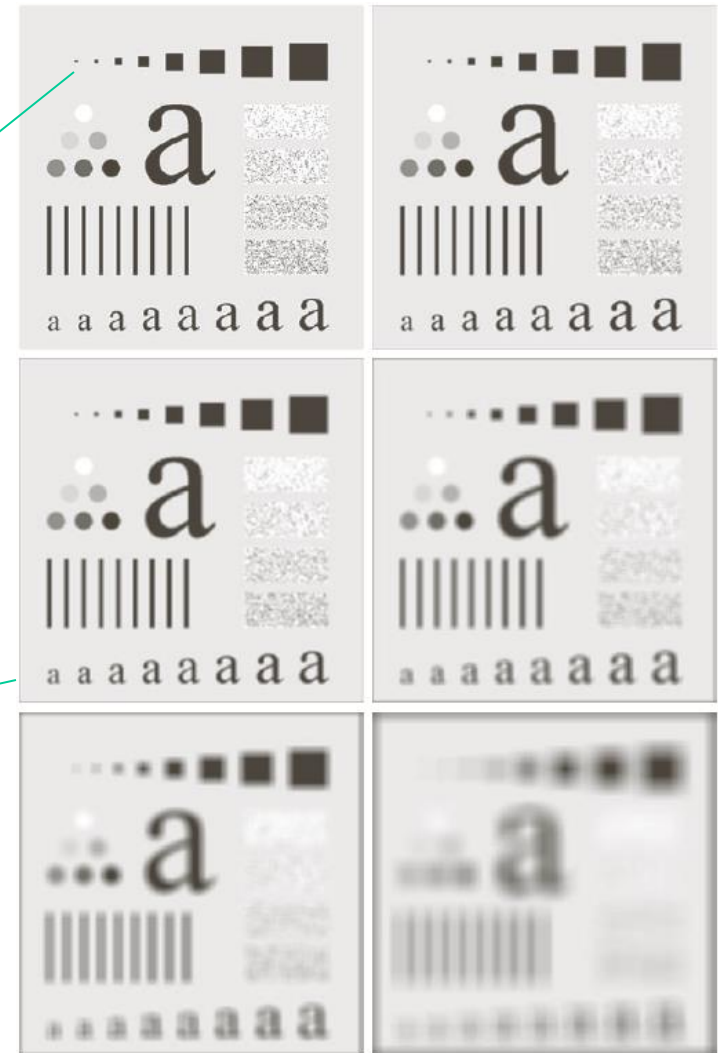
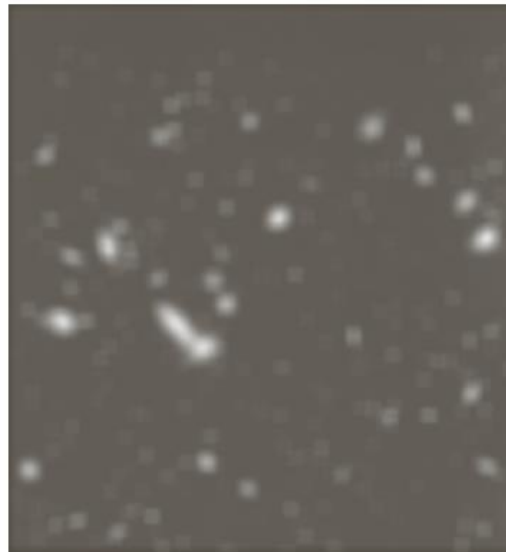
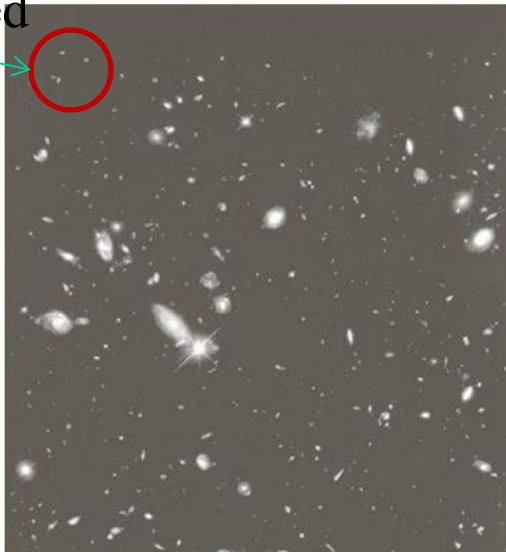
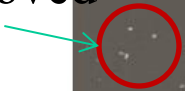


Image Smoothing and Thresholding

removed



a b c

FIGURE 3.34 (a) Image of size 528×485 pixels from the Hubble Space Telescope. (b) Image filtered with a 15×15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)

Sharpening Spatial Filters

smooth



sharpen



<http://www.bythom.com/sharpening.htm>

Sharpening – highlight the transitions in intensity by differentiation



Smoothing – blur the transitions by summation

Sharpening Spatial Filters

Sharpening – highlight the transitions in intensity by differentiation

- Electric printing
- Medical imaging
- Industrial inspection

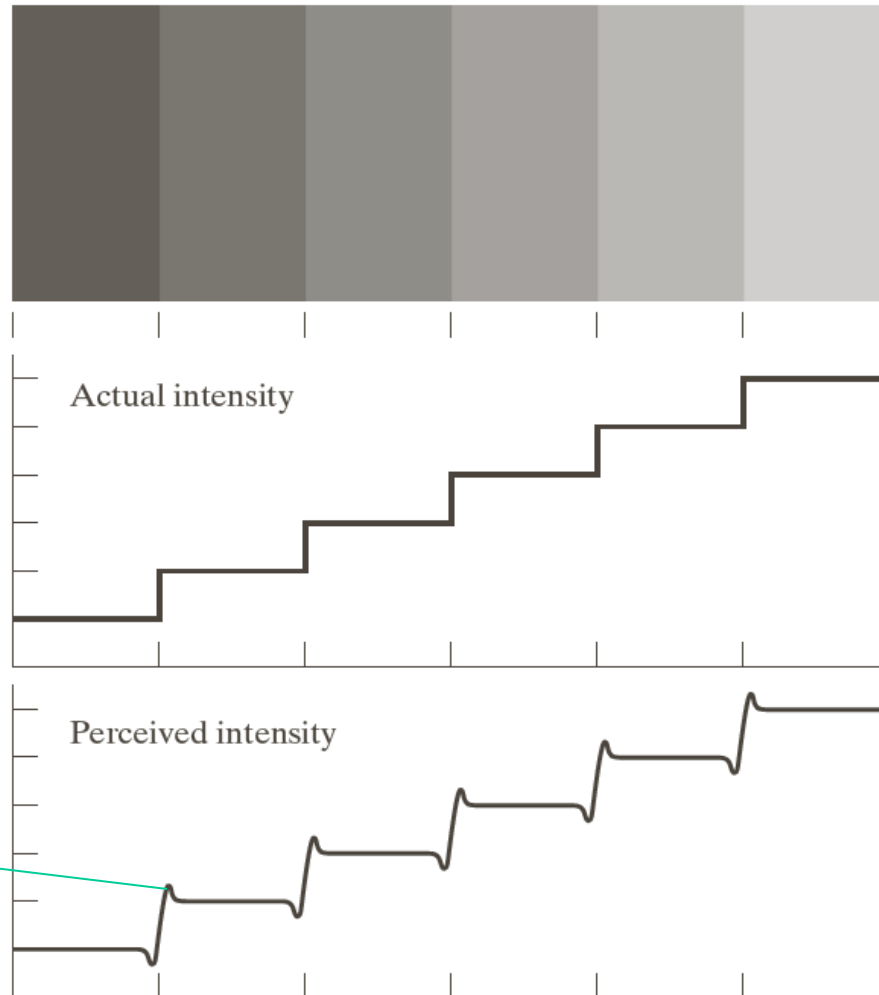
Compared to smoothing – blur the transitions by summation

Perceived Intensity is Not a Simple Function of the Actual Intensity (1)

a
b
c

FIGURE 2.7

Illustration of the Mach band effect. Perceived intensity is not a simple function of actual intensity.



Enhance/amplify difference by image sharpening

Sharpening Spatial Filters

$$g(x, y) = f(x, y) + c * e(x, y)$$

Original image

Sharpened image Magnifying factor Edge map

We will briefly introduce edge detection here and will have a more comprehensive discussion when we discuss image segmentation.

Spatial Filters for Edge Detection

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

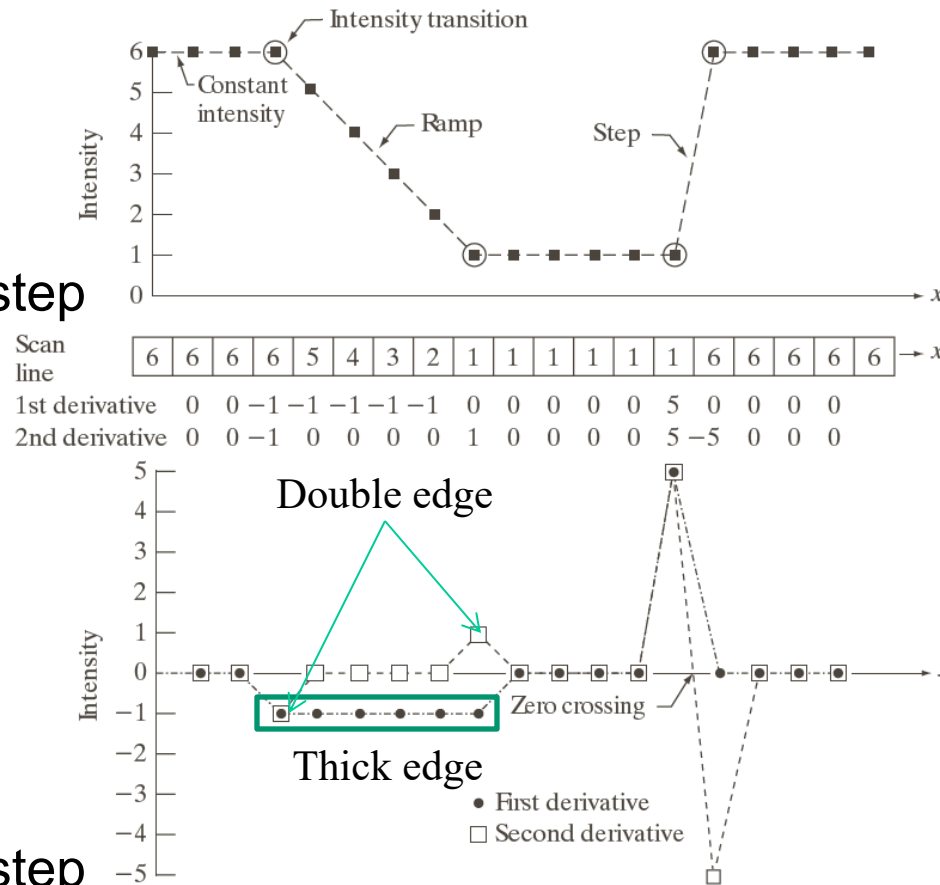
Nonzero at

- onset of ramp and step
- along ramp

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

Nonzero at

- onset of ramp and step
- end of ramp and step



a
b
c

FIGURE 3.36

Illustration of the first and second derivatives of a 1-D digital function representing a section of a horizontal intensity profile from an image. In (a) and (c) data points are joined by dashed lines as a visualization aid.

First-order VS Second-order Derivative for Edge Detection

- **First-order derivative produces thick edge along the direction of transition**
- **Second-order derivative produces thinner edges**

Gradient for Image Sharpening

Direction of change

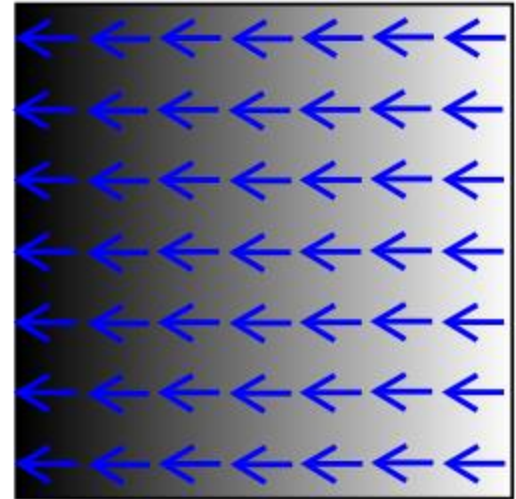
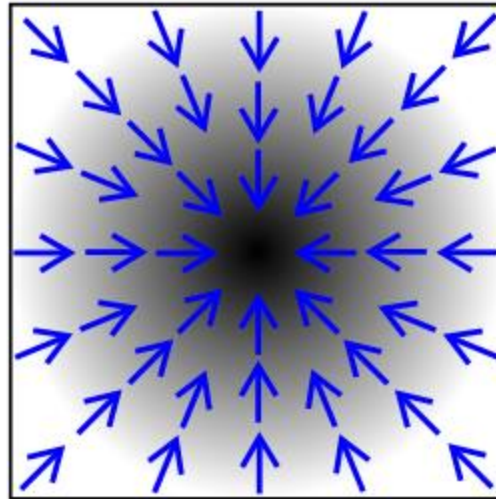
$$\nabla f = \text{grad}(f) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

Magnitude of change (gradient image)

$$M(x, y) = \text{mag}(\nabla f)$$

$$= \sqrt{g_x^2 + g_y^2}$$

$$M(x, y) \approx |g_x| + |g_y|$$



http://en.wikipedia.org/wiki/Image_gradient

Gradient for Image Sharpening

Sum of the coefficients is 0 – the response of a constant region is 0

Edge detectors:

- Roberts cross – fast while sensitive to noise
- Sobel - smooth

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

-1	0	0	-1
0	1	1	0

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

a
b c
d e

FIGURE 3.41

A 3×3 region of an image (the z s are intensity values). (b)–(c) Roberts cross gradient operators. (d)–(e) Sobel operators. All the mask coefficients sum to zero, as expected of a derivative operator.

Laplacian for Image Sharpening

2D Isotropic filters – rotation invariant

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Laplacian operator

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + 2 \frac{\partial^2 f}{\partial x \partial y}$$

$$g(x, y) = f(x, y) + c[\nabla^2 f(x, y)]$$

Image sharpening with Laplacian

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

a b
c d

FIGURE 3.37
(a) Filter mask used to implement Eq. (3.6-6).
(b) Mask used to implement an extension of this equation that includes the diagonal terms.
(c) and (d) Two other implementations of the Laplacian found frequently in practice.

Image Sharpening

Scale the Laplacian by shifting the intensity range to $[0, L-1]$



a
b c
d e

FIGURE 3.38

(a) Blurred image of the North Pole of the moon.
(b) Laplacian without scaling.
(c) Laplacian with scaling.
(d) Image sharpened using the mask in Fig. 3.37(a). (e) Result of using the mask in Fig. 3.37(b). (Original image courtesy of NASA.)

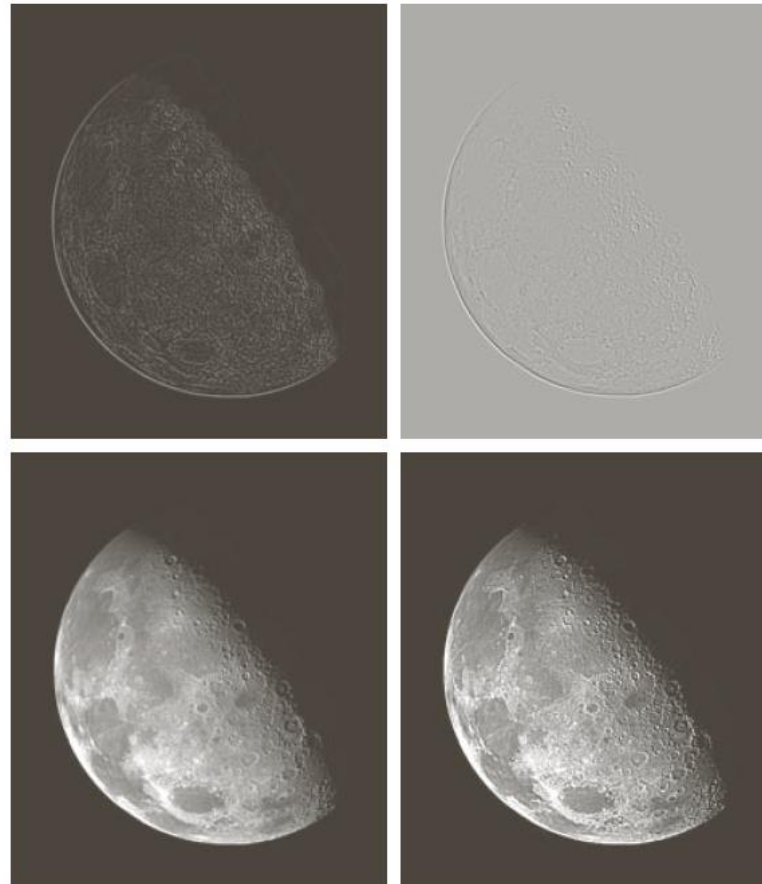


Image Sharpening by Unsharp Masking and Highboost Filtering

- 1. Blur the original image**
- 2. Subtract the blurred image from the original to get the mask**
- 3. Add the mask to the original**

$$g(x, y) = f(x, y) + k * (f(x, y) - \bar{f}(x, y))$$
$$k \geq 0$$

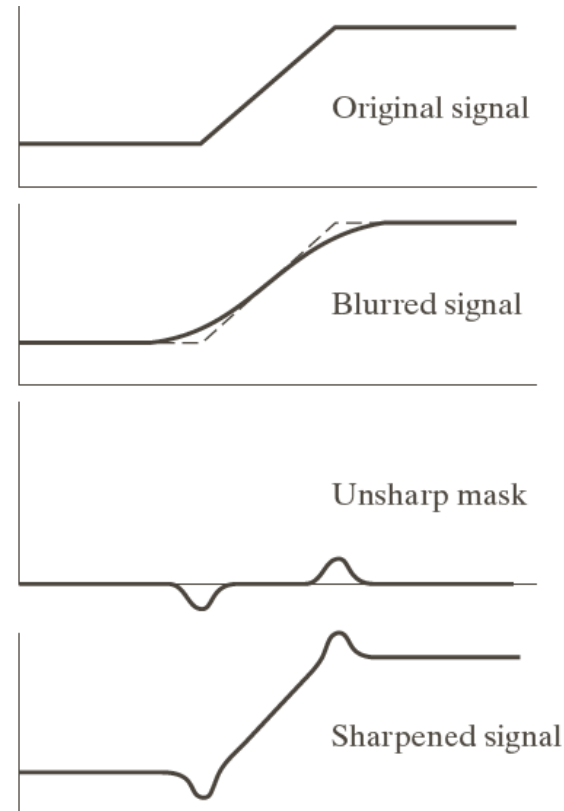


FIGURE 3.39 1-D illustration of the mechanics of unsharp masking. (a) Original signal. (b) Blurred signal with original shown dashed for reference. (c) Unsharp mask. (d) Sharpened signal, obtained by adding (c) to (a).

When $k > 1$, it becomes a highboost filtering.

An Example

Original



Blurred



Unsharp mask



Unsharp masking $k=1$



Highboost filtering $k=4.5$



a
b
c
d
e

FIGURE 3.40

(a) Original image.

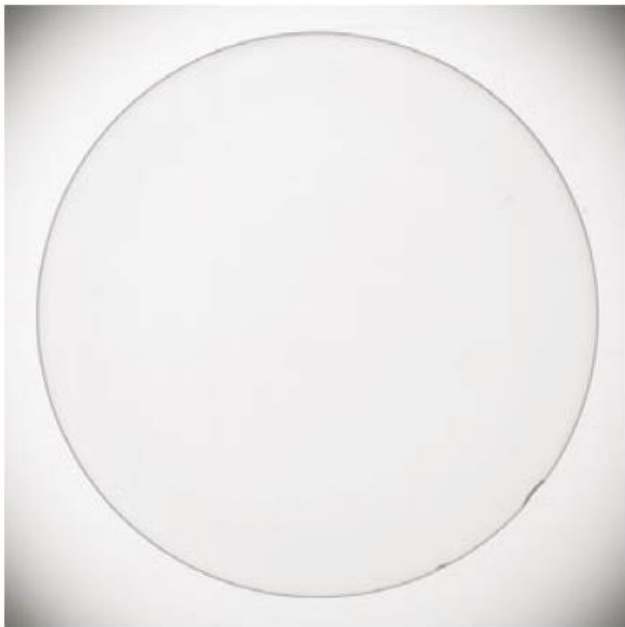
(b) Result of blurring with a Gaussian filter.

(c) Unsharp mask. (d) Result of using unsharp masking.

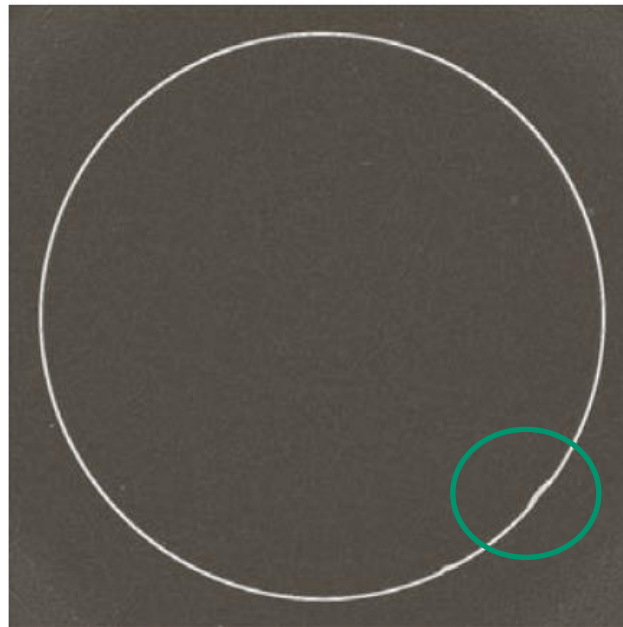
(e) Result of using highboost filtering.

Gradient for Image Sharpening -- Example

Original

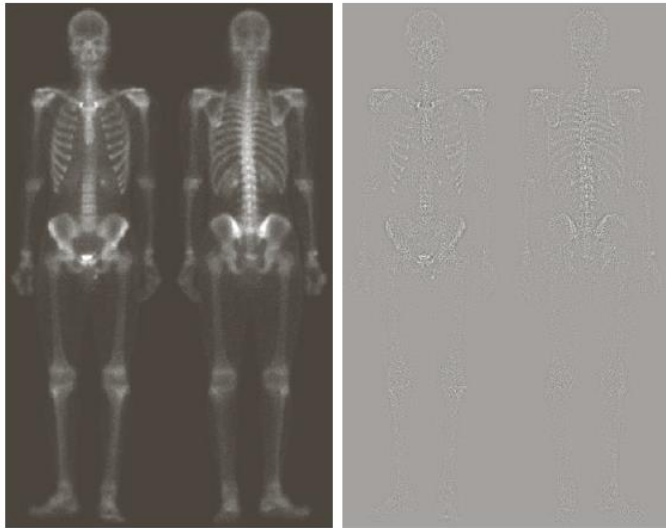


Sobel gradient



An application in industrial defect detection.

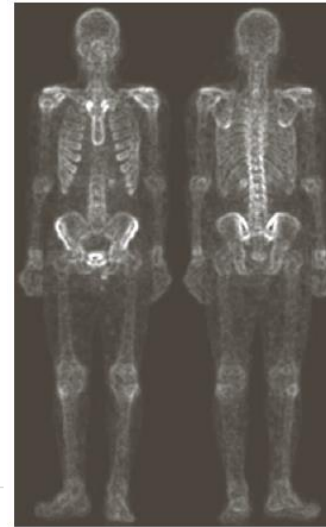
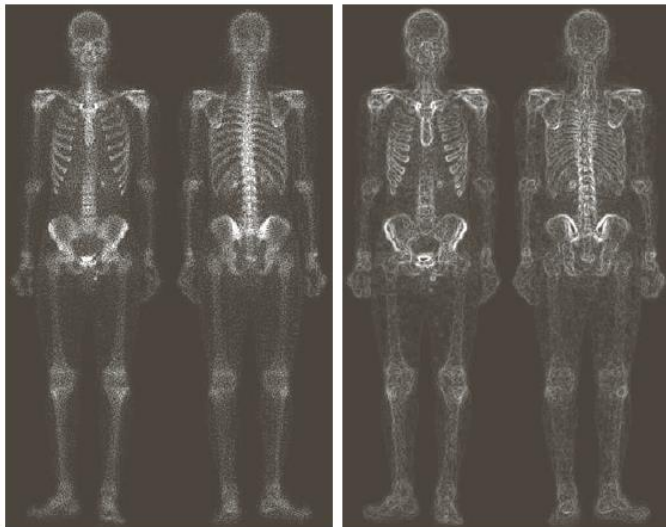
Combining Spatial Enhancement Methods



a b
c d

FIGURE 3.43

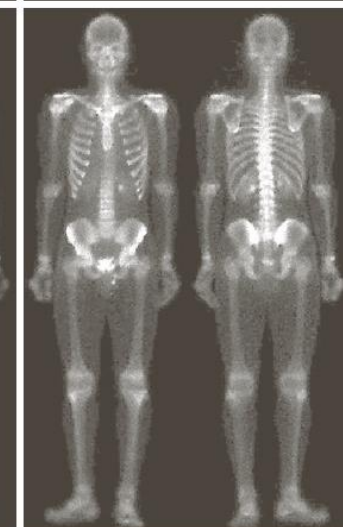
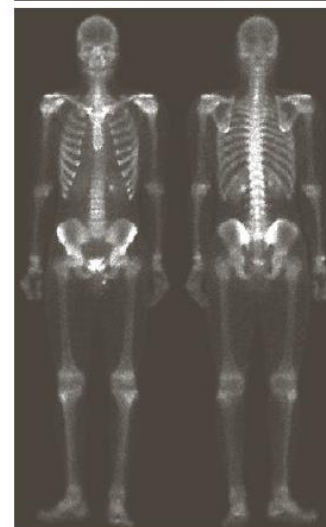
(a) Image of whole body bone scan. (b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b). (d) Sobel gradient of (a).



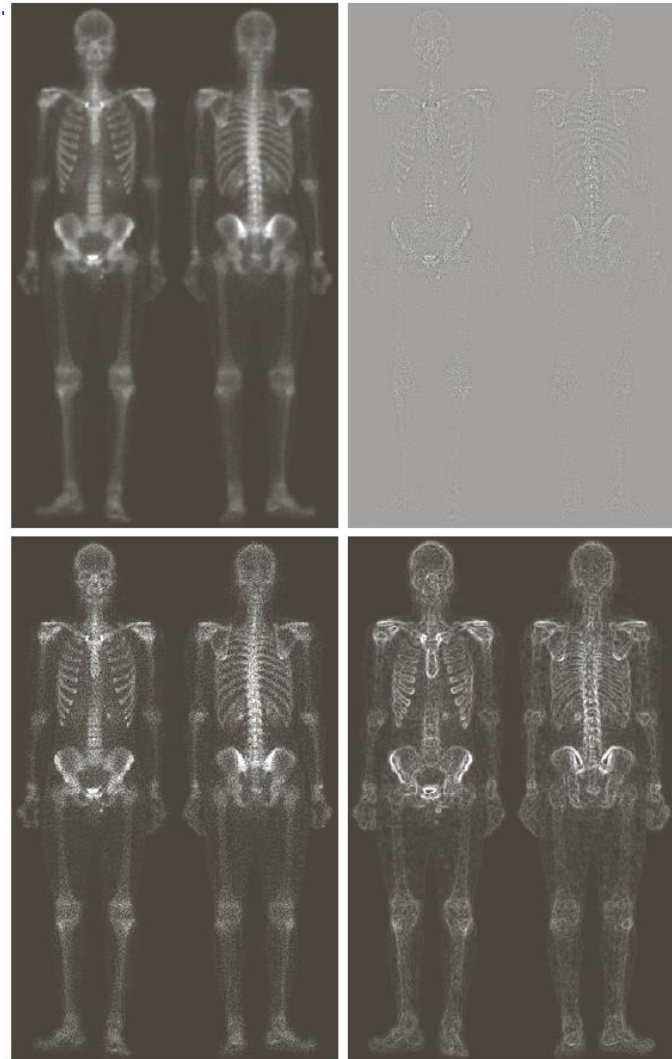
e f
g h

FIGURE 3.43
(Continued)

(e) Sobel image smoothed with a 5×5 averaging filter. (f) Mask image formed by the product of (c) and (e). (g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)



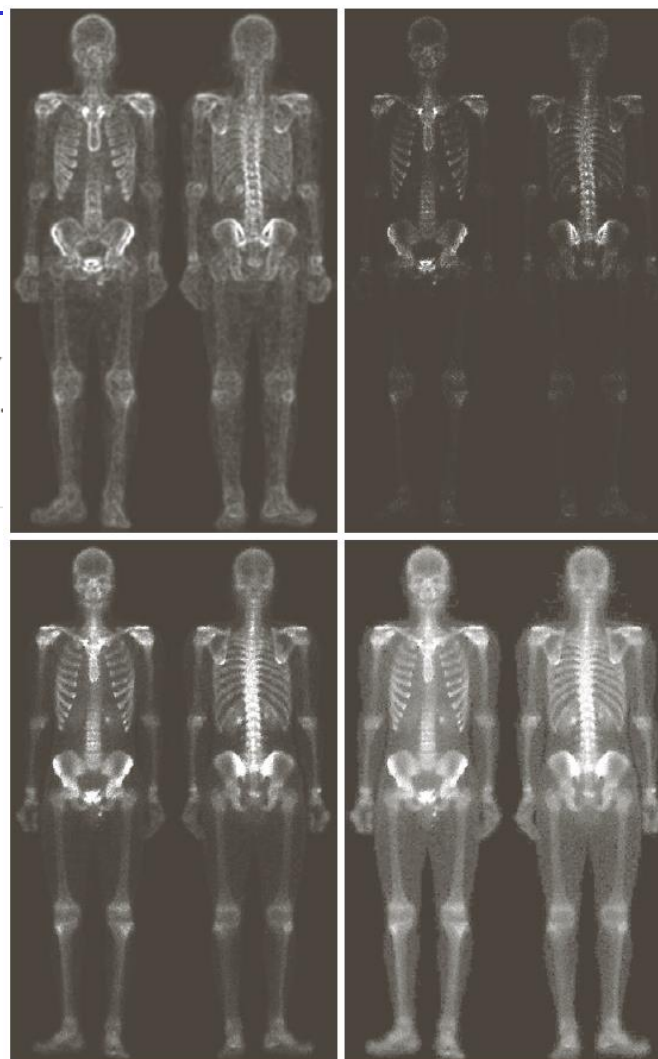
Combining Spatial Enhancement Methods



a b
c d

FIGURE 3.43

(a) Image of whole body bone scan. (b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b). (d) Sobel gradient of (a).



e f
g h

FIGURE 3.43
(Continued)

(e) Sobel image smoothed with a 5×5 averaging filter. (f) Mask image formed by the product of (c) and (e). (g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)

Order-Statistic (Nonlinear) Filtering

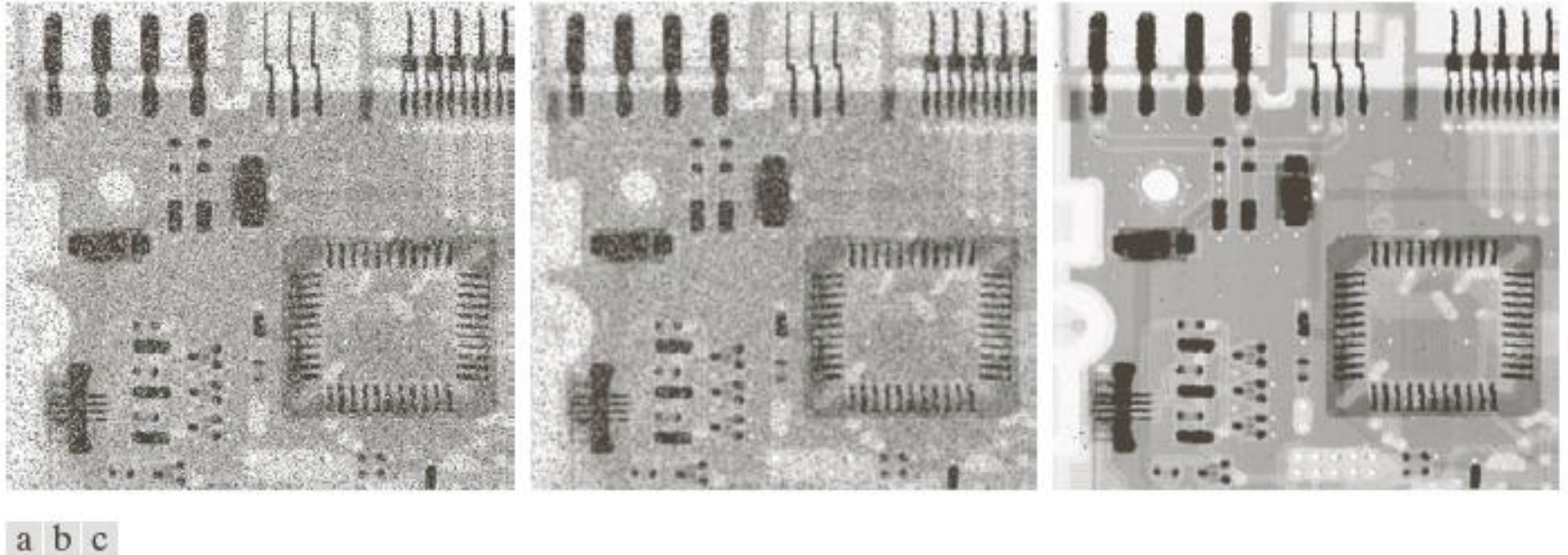


FIGURE 3.35 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3×3 averaging mask. (c) Noise reduction with a 3×3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Order-statistic filtering – rank the pixel values in the filter window and assign the center pixel according to the property of the filter

- Median
- Min/max

Reading Assignments

Next class, we will start Chapter 4: Filtering in the Frequency Domain